Appeal Clock Simulator: Initiation document · Case 03

Following chapter 1 (Initiation) of the Magshimim project-book template.

Description

A simulation of the queue of appeals against roll deletions: how many appeals arrive each day, how many each tribunal hears, how long each case takes, and when each appeal reaches a hearing. It shows in advance whether the queue will clear before the election, and what happens if tribunals are added, priorities change or hearings get shorter.

Goal

In the story, appeals do not fail; most succeed. They arrive late: in West Bengal, per a September 2026 report, 37 lakh appeals waited for 19 tribunals. An appeal won after the election does not restore the vote. Goal: turn an abstract injustice into a date, and give resource decision-makers a tool that says how many tribunals are needed, and when.

Audience

- **Primary:** the Election Commission and state high courts deciding on tribunals.
- **Secondary:** journalists and rights groups who want to show in advance that the queue will not clear.
- **Scenario:** retired Justice Hari Prasad (a fictional character) sees that at the current pace only about a quarter of appeals are heard before the election, and almost all with 40 tribunals instead of 19.

Existing solutions

- **e-Courts / National Judicial Data Grid:** shows pending cases per court; stock, not forecast.
- **Queueing models (Erlang, discrete-event simulation):** standard planning tools; tools, not a product.
- **ECI appeal reports:** cumulative counts without forecasting.

Technological core

A discrete-event queue simulation: arrivals, service per tribunal, priorities, compared with a deadline.

Main technologies

Python with SimPy; Plotly/Dash; PostgreSQL if data is obtained.

Challenges and sources

- **Data:** hearing rates, durations and success by reason are held only by the ECI, at case level.
- **Behaviour:** how many removed people even know and appeal is unknown.

Initiation check: the napkin

Before building a simulation we did the simple sum: 3,700,000 appeals ÷ (19 tribunals × 300 hearings a day) ≈ **650 working days**. Even at an optimistic 1,000 hearings per tribunal per day it is about 195 days. The answer to "will the queue clear before the election?" is "no", and it is clear without a product.

And the only user holding case-level queue data is the same central body that runs the queue. It does not need a simulation to know the queue is long. The voter, who is actually harmed, gets nothing from the tool: they still do not know they were removed.

Hardware

None.

Conclusion

❌ Falls at initiation. The initiation itself answered the question on a napkin: the answer needs no product, and precise data sits only with those who already know. **What would have saved it:** case-level queue data, and turning the tool from a view from above into something the voter receives: "you were removed, and your appeal will be heard on date X".